Operating Systems Mini Project - Review 1

Project Title: File Operation Failure Diagnoser

Course: Operating Systems

Date: August 4, 2026

Team Members:

- Yash Pradhan (24BCE0702)
- Anjini Pandey (24BCE0714)
- Dhruv Kothari (24BCE0570)

1. Problem Statement and Motivation

Problem Statement:

When users attempt routine file operations—such as deleting, moving, renaming, or copying—operating systems frequently block the action with generic and ambiguous error messages like "File is in use by another program," "Access Denied," or "Action cannot be completed." The OS fails to provide granular details regarding which specific background process, synchronization engine, or file attribute is causing the failure. Consequently, users are left to blindly guess the cause, often resorting to restarting their entire system to release hidden process locks or resolve attribute conflicts.

Motivation:

The primary motivation behind this project is to bridge the gap between complex Operating System resource management (specifically process handles and file system locks) and user-end troubleshooting. While the OS correctly enforces protection and mutual exclusion (to prevent data corruption from concurrent writes), the abstraction layer hides crucial diagnostic information from the user. By developing a dedicated diagnostic tool, we aim to demystify these errors. This project will allow us to practically apply core OS syllabus concepts—specifically Process Management, System Calls, and File Systems—to build a solution that saves time and reduces frustration for everyday computer users.

2. Objectives of the Project

The primary objective of this project is to create an intuitive desktop application that actively diagnoses and explains file operation failures. The specific objectives are:

- **Process and Handle Detection:** To identify and display the exact active process(es) holding a lock or handle on a specified file, utilizing process management libraries (e.g.,

psutil).
- **File System Attribute Analysis:** To evaluate the target file's metadata and access control rules, checking for restrictions such as "Read-Only" flags, hidden attributes, or lack of ownership permissions.
- **Synchronization Conflict Identification:** To detect if cloud storage synchronization services (like OneDrive or Google Drive) are actively accessing and thereby locking the file.
- **Actionable Resolution:** To provide users with safe, human-readable recommendations (and actionable buttons, such as "Terminate Process") to safely release the lock without requiring a system reboot.
- **User Interface Development:** To build a graphical or context-menu-based interface that allows users to easily select or drag-and-drop the problematic file for immediate diagnosis.

3. Literature Survey / Related Work

To understand the current landscape of file diagnostic tools, a survey of existing operating system utilities, architectural literature, and third-party software was conducted. The findings highlight both the capabilities and limitations of current solutions in addressing concurrent file access and resource management issues.

3.1. Architectural Foundations and Resource Management

- **OS Design and File Systems:** According to standard operating system texts such as *Operating Systems: Principles and Practice* [1] and *Operating System Concepts* [2], low-level process management and file systems are typically implemented in the kernel. Modern operating systems manage concurrent access to files by utilizing system calls to assign file handles and lock mechanisms. This ensures data integrity by preventing overlapping write operations from multiple processes simultaneously. However, this robust abstraction, managed by mechanisms like the Windows Object Manager, often results in generic error messages for the user when a lock is encountered [3].
- **The Synchronization Challenge:** Contemporary workflows often involve background synchronization engines (like OneDrive or Google Drive). These engines continuously poll and lock files for uploading/downloading, which frequently leads to concurrent execution conflicts where a user is locked out of their own file during a sync state [4].

3.2. Diagnostic and Developer Utilities

- **Microsoft Sysinternals - Handle and Process Explorer:**
  - *Overview:* Developed by Microsoft, these tools interact directly with the OS. The Handle command-line utility displays open handles for any process, while Process Explorer provides a Graphical User Interface (GUI) equivalent to track active processes and their associated file handles [5].
  - *Limitations:* While technically robust, Process Explorer is designed for developers and system administrators. Identifying a locked file requires manually using a specific "Find Handle or DLL" search function. It is not designed for quick diagnostics by an average user and completely ignores non-process-related failures (e.g., simple file permission

issues).

3.3. Consumer-Facing Shell Extensions

- **Microsoft PowerToys - File Locksmith:**
  - *Overview:* A utility integrated into the Windows shell that allows users to right-click a file and view a list of processes currently utilizing it [6].
  - *Limitations:* File Locksmith is heavily reliant on process detection. If an operation fails due to read-only attributes, misconfigured ownership, or complex cloud-sync delays that are not registered as standard active processes, File Locksmith often reports "No processes found," providing no further assistance to the user.

3.4. Third-Party Unlocking Tools

- **Aggressive Terminators (e.g., LockHunter, IObit Unlocker):**
  - *Overview:* These tools are popular free utilities designed to integrate into the Windows context menu. They identify processes holding locks on specific files or folders and allow users to aggressively terminate those processes, rename, move, or delete the file [7][8].
  - *Limitations:* The primary design philosophy of these tools is brute-force termination. They lack diagnostic feedback to educate the user on *why* the file was locked. Furthermore, force-killing processes without understanding their origin can lead to system instability, data corruption, or the disruption of critical background services.

3.5. Feature Comparison Summary

The following table summarizes the capabilities of the reviewed tools compared to the proposed File Operation Failure Diagnoser:

Feature	Process Explorer	File Locksmith	LockHunter	Proposed System
Detect Process Lock	✓	✓	✓	✓
Permission Analysis	X	X	X	✓
File Attribute Analysis	X	X	X	✓
Sync Activity Check	X	X	X	✓ (Basic)
Human-reada	X	X	Limited	✓

ble Diagnosis				

3.6. Conclusion of Survey

The review of existing literature and software reveals a gap in the market. Current tools fall into three categories: complex developer utilities (Process Explorer), limited process-only checkers (File Locksmith), or aggressive executioners (LockHunter, IObit Unlocker). Our proposed File Operation Failure Diagnoser aims to fill this gap by providing a holistic diagnostic approach—evaluating locks, attributes, permissions, and sync states—while presenting the root cause in an educational, user-friendly manner rather than immediately resorting to brute-force process termination.

4. References

[1] T. Anderson and M. Dahlin, *Operating Systems: Principles and Practice*, 2nd ed. Recursive Books, 2014.

[2] A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 10th ed. Hoboken, NJ: Wiley, 2018.

[3] M. E. Russinovich, D. A. Solomon, and A. Ionescu, *Windows Internals, Part 1*, 6th ed. Redmond, WA: Microsoft Press, 2012.

[4] "Concurrent file access," Server Fault, Oct. 25, 2011. [Online]. Available: https://serverfault.com/questions/324507/how-do-concurrent-filesystems-work.

[5] Microsoft Corporation, "Handle - Sysinternals," *Microsoft Learn*. [Online]. Available: https://learn.microsoft.com/en-us/sysinternals/downloads/handle

[6] Microsoft Corporation, "PowerToys File Locksmith utility," *Microsoft Learn*. [Online]. Available: https://learn.microsoft.com/en-us/windows/powertoys/file-locksmith

[7] "Best Drive Unlocker for Windows 11," Ithy, Mar. 06, 2025. [Online]. Available: https://ithy.com/article/drive-unlocker-windows-s81lw5g8

[8] Crystal Rich Ltd., "LockHunter is a free 64/32 bit tool to delete files blocked by any processes," *LockHunter*. [Online]. Available: https://lockhunter.com/

OPERATING SYSTEMS

FILE OPERATION FAILURE DIAGNOSER

Operating Systems Mini Project – Review 1

Course

Operating Systems

Date

August 4, 2026

Team Members

- Dhruv Kothari (24BCE0570)
- Yash Pradhan (24BCE0702)
- Anjini Pandey (24BCE0714)

Problem Statement

Existing Issues

- Users frequently encounter generic file errors: **“Access Denied”**, **“File is in use”**, **“Action cannot be completed”**.
- Operating systems surface error codes, but rarely explain the precise cause.
- Users often reboot or guess fixes without identifying the root cause, causing downtime and data risk.

Why This Problem Occurs

- Permission conflicts between users, applications, and file ownership.
- A file may be **locked by another process**, backup service, or sync client.
- Corrupted metadata, path issues, or hidden system restrictions make the true cause harder to identify.

Real-Life Examples

- Saving a document in **OneDrive** while it is being accessed by another app.
- Trying to delete a folder that is still open in File Explorer or being scanned by antivirus software.
- Moving a file from an external drive when the drive has a temporary read/write conflict.

Impact on Users

- They do not know which application is blocking the file.
- They waste time trying random fixes that may not work.
- They may lose unsaved work or interrupt important workflows.
- They feel frustrated because the message is vague and non-actionable.

Motivation



Bridge OS & User

Translate low-level resource state into **actionable user guidance**.



Apply Core Concepts

Leverage **process management**, **system calls**, and **filesystem internals**.



Speed Troubleshooting

Reduce unnecessary reboots and **unsafe interventions**.

Why This Project Matters

- Helps end users and IT professionals understand errors faster and choose the right next step.
- Applies core OS concepts in a practical setting, including **process management**, file systems, and permissions.
- Improves real-world troubleshooting by reducing guesswork, downtime, and risky workarounds.
- Strengthens academic learning by connecting theory to observable system behavior and diagnostics.

Project Importance

- Provides clearer answers when users encounter **generic file errors**.
- Helps identify the real cause behind **permission** and **file-lock** issues.
- Reduces time lost to trial-and-error fixes and unnecessary restarts.
- Builds practical troubleshooting skills that apply to everyday operating systems.

Key Benefits

User Confidence

Clear explanations help users resolve issues without unnecessary stress.

Faster Troubleshooting

More precise root-cause guidance shortens diagnosis time and reduces downtime.

Educational Value

Turns operating system theory into a hands-on learning experience.

Objectives

Detect processes locking a file

Identify the PID, process name, and handle type so users can quickly find what is preventing access.

Detect cloud-sync activity

Spot OneDrive and Google Drive interference before users make risky changes to a synced file.

Analyze permissions and ownership

Check file permissions and ownership to explain missing privileges and clarify why access is being denied.

Provide safe recommendations

Deliver human-readable guidance that prioritises safety and helps users choose the least disruptive next step.

Check file attributes

Review read-only and hidden attributes to surface how they affect editing, saving, and visibility.

Deliver a simple desktop app

Create a clear interface for end users so technical details are easy to understand and act on.

Literature Survey — Existing Tools

Microsoft Process Explorer

Overview: A detailed diagnostic tool for viewing running processes and their relationships.

- Advantages: Detailed handle view and process tree.
- Limitations: Technical UI; limited explanation for novices.

Key Takeaway: Best for advanced users, not suitable for general users.

Microsoft File Locksmith

Overview: A built-in utility for quickly identifying what is locking a file.

- Advantages: Identifies locking handles quickly.
- Limitations: Minimal guidance on permissions or cloud sync.

Key Takeaway: Good for basic lock detection but lacks context.

LockHunter

Overview: A file-unlocking tool designed to help users remove file locks and regain access.

- Advantages: Simple unblock options and file unlock actions.
- Limitations: Forceful actions can risk data; lacks contextual recommendations.

Key Takeaway: Effective but potentially risky without proper guidance.

Feature Comparison

Key capabilities evaluated across tools and our proposed system.

System / Feature	Detect Lock	Permissions	Attributes	Cloud Sync	Readable Diagnosis
Process Explorer	✓	Partial	✗	✗	Technical
LockHunter	✓	✗	Partial	✗	Action-oriented
File Locksmith	✓	✗	✗	✗	Technical
Proposed Diagnoser	✓	✓	✓	✓	✓

The Proposed Diagnoser stands out by combining lock detection, permission checks, attribute analysis, and cloud sync diagnosis into one clear, user-friendly explanation.

Proposed Solution — Overview

User selects file

Process Lock Detection

Permission Analysis

File Attribute Check

Cloud Sync Detection

Generate Diagnosis

Provide Safe Recommendations

The Diagnoser follows a clear, lightweight workflow: it captures the file context, checks for lock and permission issues, analyzes file and cloud sync attributes, then returns a prioritized diagnosis with safe next steps.

UI & Interaction

The Diagnoser uses a clean, single-window workflow to surface what matters most at a glance.

Key Interface Components

Status Panel

- Shows the file path and error type
- Lets the user know what is being diagnosed

Sync Detection Badge

- Highlights OneDrive or Google Drive
- Shows live activity status

Process List

- Displays PID, process name, and user
- Pinpoints which app is holding the file

Recommendation Panel

- Offers safe action buttons
- Request to close, Retry, or Pause sync

Permission Panel

- Summarizes owner and ACL details
- Explains access denial reasons

User Workflow & Benefits

User Workflow

User Workflow

Select Locked File The user starts by choosing the file that cannot be opened, moved, or edited.

Review Live Diagnostics The interface reveals the process list, permission details, and sync status in one place.

Choose Safe Recommendation The user applies a recommended next step without needing to guess or force risky changes.

Confirm Resolution The app rechecks access and confirms whether the file is now available.

Benefits of This Interface

Faster Diagnosis

Users can identify the cause of the lock quickly without switching between tools.

Clearer Context

Process, permission, and sync signals are grouped together for easy comparison.

Safer Decisions

Action buttons focus on low-risk remedies instead of abrupt or destructive fixes.

Better User Experience

A single-window layout keeps the workflow simple and reduces cognitive load.

Expected Outcome

Technical Benefits

- More efficient **process inspection** by surfacing PID, handle type, and ownership details in one view.
- Better **permission analysis** through clearer visibility into ACL differences and required privilege level.
- Improved **sync detection** for identifying cloud-backed files and live activity states more accurately.
- Safer remediation by prioritizing non-destructive actions before escalation or termination.

User Benefits

- Clearer diagnoses reduce confusion and help users understand why a file is locked.
- **Safe actions** such as retrying or pausing sync reduce the need for risky workarounds.
- Less interruption from avoiding unnecessary reboots and force-unlocks.
- A more intuitive workflow makes file recovery feel faster and more dependable.

Educational Benefits

- Turns abstract **Operating Systems** concepts into a practical desktop tool.
- Demonstrates how processes, permissions, and synchronization interact in real systems.
- Supports learning through direct observation of system state and recommended actions.

Future Impact

- Can be extended to support more **sync providers** and lock sources.
- Could evolve into a broader diagnostic assistant for file access and permission issues.
- Provides a foundation for smarter automation, reporting, and enterprise troubleshooting workflows.

Implementation Roadmap



Implementation

- Develop kernel-edge queries for process and file state inspection
- Build user-space agent for permission and sync analysis
- Create UI front-end with a clear, actionable interface
- Target: Windows reference implementation with potential for cross-platform extension



Testing

- Instrument test scenarios: locked files, permission errors, cloud sync conflicts
- Measure time-to-resolution and accuracy of diagnoses
- Validate safety of recommended actions
- Gather feedback from test users



Faculty Demo

- Live demonstration: reproduce common failures and show stepwise recommendations
- Explain how the tool applies Operating Systems concepts
- Answer questions about design decisions and implementation details
- Show the complete workflow from file selection to resolution



Deliverables

- Source code with clear documentation and comments
- Comprehensive test suite covering edge cases
- User manual with screenshots and troubleshooting guide
- Short video walkthrough demonstrating the tool in action

We will prioritise safety-first recommendations and clear, teachable outputs for the faculty review.

Problem

File operation failures are common, but users often receive unclear explanations.

This can lead to confusion, repeated attempts, and unsafe decisions.

Solution

A desktop diagnoser identifies likely causes by analyzing processes, permissions, file attributes, and cloud sync activity.

It then provides safe, human-readable recommendations.

Key Features

- Process lock detection with PID and handle type identification
- Permission and ownership analysis with ACL comparison
- Cloud sync detection, file attribute inspection, and safe action recommendations

Expected Impact

Faster troubleshooting, reduced downtime, and lower data risk for users.

A clearer workflow also demonstrates key Operating Systems concepts in practice.

This project demonstrates practical application of Operating Systems concepts while solving a real-world user problem.

THANK YOU

Questions & Discussion

Thank you for your time and valuable feedback.

Operating Systems Mini Project - File Operation Failure Diagnoser

Team Members:

- Dhruv Kothari (24BCE0570)
- Yash Pradhan (24BCE0702)
- Anjini Pandey (24BCE0714)